



Relationship between technology readiness, AI adoption and value creation in hospitality industry: Moderating role of technological turbulence

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ABSTRACT

This study explores the relationship between technology readiness, AI adoption, dynamic capabilities, sustainable competitive advantage, and technological turbulence, assessing their collective impact on value creation within the Malaysian hospitality industry. Employing the Resource-Based View, Dynamic Capabilities Framework, and Contingency Theory, the research analyzes these strategic elements' roles in enhancing organizational performance. Using stratified purposive sampling, data was collected from 336 respondents. Structural equation modeling was utilized to analyze the relationships among the constructs. The results reveal that technology readiness and AI adoption positively influence value creation, with AI adoption playing a direct role in enhancing organizational capabilities and efficiencies. Interestingly, dynamic capabilities exhibited a complex impact, initially impeding value creation due to transitional challenges, but ultimately proving crucial in navigating technological turbulence. Sustainable competitive advantage emerged as a significant mediator, translating AI adoption and dynamic capabilities into enhanced value creation. However, the moderating effects of technological turbulence were selective, significantly influencing only the relationship between dynamic capabilities and value creation. These findings underscore the nuanced interplay between strategic resources and environmental factors in driving value creation in a technologically evolving industry.

1. Introduction

The hospitality industry stands at a pivotal juncture, melding traditional service values with the transformative potential of artificial intelligence (AI). As AI emerges as a crucial force, it drives unprecedented enhancements in service quality, operational efficiency, and personalized guest (Bulchand-Gidumal et al., 2023; Guo et al., 2024). However, the integration of AI demands a delicate balance of technology readiness and continuous adaptation to evolving market demands—areas that remain critically underexplored (Parvez, 2020; Warner and Wäger, 2019).

Recent developments underscore the urgency of addressing this balance. For example, Marriott International has been at the forefront of implementing AI-driven chatbots to handle customer inquiries, significantly enhancing response times and customer satisfaction (Al-Hyari et al., 2023). Similarly, Hilton has deployed AI-based systems for predictive maintenance in their properties (Takyar, 2024), resulting in a

notable reduction in operational disruptions and maintenance costs. These examples illustrate how AI is not only meeting the growing expectation for hyper-personalization but also establishing new benchmarks for service excellence in the industry. As reported by *The Wall Street Journal* and *CNBC* in 2023, such AI-driven innovations are increasingly becoming the standard in the competitive hospitality market (Captain, 2023; Siu, 2023).

Moreover, advancements in machine learning, a key subset of AI, have significantly improved data quality and algorithmic accuracy (Okafor et al., 2020; Whang et al., 2023). Despite these technological strides, the sector faces significant challenges in applying AI to enhance customer interactions and streamline operations (Bassano et al., 2020). For instance, the ethical implications of AI deployment, particularly regarding data privacy, have become a focal point of concern. Reports from *The Deloitte* reveal growing unease over AI's role in guest monitoring and data usage, sparking debates on how hospitality businesses can responsibly integrate these technologies without compromising

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